EM waves in vacuum

$$\begin{cases} \vec{\nabla}\cdot\vec{E}=0 & \vec{\nabla}\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \\ \vec{\nabla}\cdot\vec{B}=0 & \vec{\nabla}\times\vec{B}=\mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t} \end{cases}$$

rule 8

$$\vec{\nabla}\times\vec{\nabla}\times\vec{E} = \nabla\underbrace{(\vec{\nabla}\cdot\vec{E})}_{\phi} - \nabla^2\vec{E} = -\frac{\partial}{\partial t}\vec{\nabla}\times\vec{B} = -\mu_0\varepsilon_0\frac{\partial^2\vec{E}}{\partial t^2}$$

$$\vec{\nabla}\times\vec{\nabla}\times\vec{B} = \nabla\underbrace{(\vec{\nabla}\cdot\vec{B})}_{\phi} - \nabla^2\vec{B} = \mu_0\varepsilon_0\frac{\partial}{\partial t}\vec{\nabla}\times\vec{E} = -\mu_0\varepsilon_0\frac{\partial^2\vec{B}}{\partial t^2}$$

Both satisfy <u>wave equation</u>

$$\nabla^2\vec{E} = \frac{1}{c^2}\frac{\partial^2\vec{E}}{\partial t^2}, \quad c^2=\frac{1}{\varepsilon_0\mu_0}$$

$$c=\frac{1}{\sqrt{\varepsilon_0\mu_0}}=3.00\cdot10^8\,\mathrm{m/s}\,!$$

<u>Plane waves:</u>

$$\left.\begin{aligned}\tilde{E}(z,t)&=\tilde{E}_0\,e^{i(kz-\omega t)}\\ \tilde{B}(z,t)&=\tilde{B}_0\,e^{i(kz-\omega t)}\end{aligned}\right\} \quad -k^2\tilde{E}=-\frac{\omega^2}{c^2}\tilde{E}$$

$$\omega^2=c^2k^2$$

$$\boxed{\omega=ck}$$

physical E and B are real parts of $\tilde{E}$ and $\tilde{B}$.

$$\vec{\nabla}\cdot\tilde{E} = \frac{\partial}{\partial x}\tilde{E}_x+\frac{\partial}{\partial y}\tilde{E}_y+\frac{\partial}{\partial z}\tilde{E}_z = ik\tilde{E}_z=0$$

$\vec{\nabla}\cdot\tilde{B}=0 \Rightarrow k\tilde{B}_z=0 \Rightarrow$ EM waves <u>are transverse</u>

$$\vec{\nabla}\times\tilde{E} = \begin{vmatrix}\hat{x} & \hat{y} & \hat{z}\\ \partial_x & \partial_y & \partial_z\\ \tilde{E}_x & \tilde{E}_y & 0\end{vmatrix} = \hat{x}\left(-\frac{\partial}{\partial z}\tilde{E}_y\right)+\hat{y}\left(\frac{\partial}{\partial z}\tilde{E}_x\right)+$$

$$+\hat{z}\left(\frac{\partial}{\partial x}\tilde{E}_y-\frac{\partial}{\partial y}\tilde{E}_x\right)=$$

$$=-\frac{\partial}{\partial t}\left(\hat{x}\tilde{B}_x+\hat{y}\tilde{B}_y\right)$$

$$\Rightarrow -ik\tilde{E}_y = i\omega\tilde{B}_x, \quad ik\tilde{E}_x = i\omega\tilde{B}_x$$

$$\Rightarrow \tilde{B}=\frac{k}{\omega}(\hat{z}\times\tilde{E}) \Rightarrow \tilde{B}_0=\frac{1}{c}\hat{z}\times\tilde{E}_0\,;\ |\tilde{B}_0|=\frac{1}{c}|\tilde{E}_0|$$

Maxwell's eqns in dielectrics

$$\begin{cases} \vec{\nabla}\cdot\vec{D} = \rho_f \\ \vec{\nabla}\cdot\vec{B} = 0 \end{cases} \qquad \begin{aligned} \vec{\nabla}\times\vec{E} &= -\frac{\partial\vec{B}}{\partial t} \\ \vec{\nabla}\times\vec{H} &= \vec{J}_f + \frac{\partial\vec{D}}{\partial t} \end{aligned}$$

$$\begin{cases} \vec{H} = \frac{1}{\mu}\vec{B} \\ \vec{M} = \chi_m \vec{H} \end{cases} \qquad \begin{cases} \vec{P} = \varepsilon_0 \chi_e \vec{E} \\ \vec{D} = \varepsilon \vec{E} \end{cases} \qquad \begin{aligned} \mu &= \mu_0(1+\chi_m) \\ \varepsilon &= \varepsilon_0(1+\chi_e) \end{aligned}$$

With $\rho_f = 0$ and $J_f = 0$:

$$\begin{cases} \vec{\nabla}\cdot\vec{D} = \vec{\nabla}\cdot\varepsilon\vec{E} = \varepsilon\,\vec{\nabla}\cdot\vec{E} = 0 \\ \vec{\nabla}\cdot\vec{B} = 0 \\ \vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \\ \frac{1}{\mu}\vec{\nabla}\times\vec{B} = \varepsilon\frac{\partial\vec{E}}{\partial t} \Rightarrow \vec{\nabla}\times\vec{B} = \varepsilon\mu\frac{\partial\vec{E}}{\partial t} \end{cases}$$

while in vacuum the only change is $\underline{\varepsilon\mu \to \varepsilon_0\mu_0}$!

But wave equation in vacuum : $\nabla^2\vec{E} = \mu_0\varepsilon_0\frac{\partial^2\vec{E}}{\partial t^2}$

$$\Rightarrow c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$$

$\Rightarrow$ in dielectric medium $\nabla^2 E = \mu\varepsilon\frac{\partial^2 E}{\partial t^2}$

$$\Rightarrow v = \frac{1}{\sqrt{\mu\varepsilon}} = \sqrt{\frac{\mu_0\varepsilon_0}{\mu_0\varepsilon_0\sqrt{\mu\varepsilon}}} = \frac{c}{n}$$

where $n = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}}$ index of refraction!

$n \geq 1 \Rightarrow v \leq c$.

$$\vec{E}(\vec{r},t) = \tilde{E}_0 \, e^{i(\vec{k}\vec{r}-\omega t)}$$

$$\vec{B}(\vec{r},\vec{t}) = \tilde{B}_0 \, e^{i(\vec{k}\vec{r}-\omega t)}$$

1) $\vec{\nabla}\cdot\vec{E} = \vec{\nabla}\cdot\vec{B} = 0 \Rightarrow i\vec{k}\cdot\tilde{E}_0 = 0, \; \vec{k}\cdot\tilde{B}_0 = 0$

2) from $\bar{\nabla}\times\bar{E} = -\frac{\partial \bar{B}}{\partial t}$

$$i\vec{k}\times\tilde{E}_0 = i\omega\tilde{B}_0 \Rightarrow \tilde{B}_0 = \frac{\vec{k}\times\tilde{E}_0}{\omega} = \frac{1}{v}\hat{k}\times\tilde{E}_0$$

↳ unit vector in direction of propagation.

$$\omega = kv$$

Note: $|\tilde{B}_0| = \frac{1}{v}|\tilde{E}_0| \Rightarrow$ mag. field's amplitude is smaller by a factor of $\frac{1}{c}$!

$\Rightarrow$ let $\tilde{E}_0 = |\tilde{E}_0|\hat{n}$

$$\Rightarrow \begin{cases} \vec{E}(r,t) = \tilde{E}_0 \, e^{i(\bar{k}\bar{r}-\omega t)} \, \hat{n} \; (\leftarrow \text{polarization vector}) \\ \vec{B}(r,t) = \frac{1}{v}(\hat{k}\times\hat{n}) \, \tilde{E}_0 \, e^{i(\bar{k}\bar{r}-\omega t)} \end{cases}$$

Energy and momentum: $u = \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{B^2}{\mu_0}\right) = \varepsilon_0 E^2$

$$B^2 = \frac{1}{c^2}E^2 = \mu_0\varepsilon_0 E^2 \Rightarrow \frac{1}{\mu_0}B^2 = \varepsilon_0 E^2$$

Pointing $\quad \vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B} = \frac{1}{\mu_0}\frac{1}{c}E_0^2\cos^2(kr-\omega t)\,\hat{k}$

$$\hat{n}\times(\hat{k}\times\hat{n}) = \hat{k}(\hat{n}\cdot\hat{n}) - \hat{n}(\hat{n}\cdot\hat{k}) = \hat{k}$$

in medium
$\vec{S} = \hat{k}\, v\, \varepsilon\, \tilde{E}_0^2\cos^2$

$$\frac{1}{\mu_0 c} = \sqrt{\frac{\varepsilon_0}{\mu_0}}\,\frac{c}{c} = \varepsilon_0 c$$

$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$$

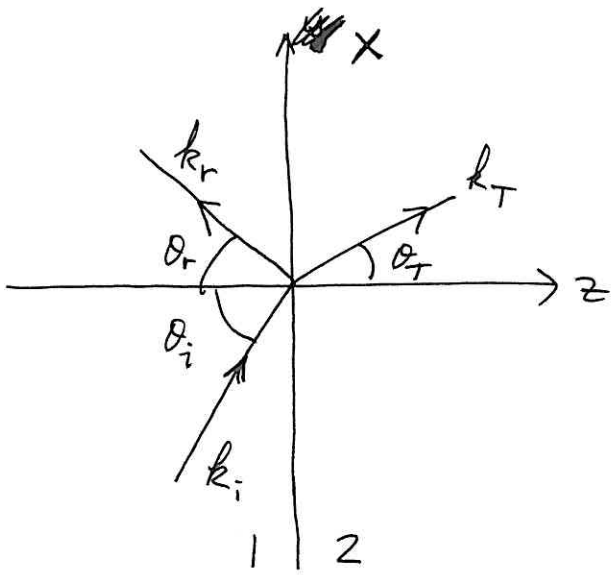
$$\frac{1}{c} = \sqrt{\mu_0\varepsilon_0}$$

$$\Rightarrow \vec{S} = \hat{k}\, c\, \varepsilon_0 E_0^2\cos^2(kr-\omega t) = \hat{k}\, c\, u$$

Boundary conditions: $\vec{D}_\perp$ continuous

$$\begin{cases} 1) & \varepsilon_1 E_1^{\perp} = \varepsilon_2 E_2^{\perp} \\ 2) & \vec{B}_\perp \text{ continuous: } \quad B_1^{\perp} = B_2^{\perp} \\ 3) & E_1^{\parallel} = E_2^{\parallel} \\ 4) & H_1^{\parallel} = H_2^{\parallel} \Rightarrow \frac{1}{\mu_1} B_1^{\parallel} = \frac{1}{\mu_2} B_2^{\parallel} \end{cases}$$

Reflection and transmission



incoming $\quad \vec{E}_i = \hat{E}_{0i}\, e^{i(\bar{k}_i \bar{r} - \omega t)} \quad \vec{B}_i = \frac{1}{v_1} \hat{k}_i \times \vec{E}_i$

reflected $\quad \vec{E}_R = \tilde{E}_{0R}\, e^{i(\bar{k}_R \bar{r} - \omega t)} \quad \vec{B}_R = \frac{1}{v_1} \hat{k}_R \times \vec{E}_R$

$\vec{E}_T = \tilde{E}_{0T}\, e^{i(\bar{k}_T \bar{r} - \omega t)} \quad \vec{B}_T = \frac{1}{v_2} \hat{k}_T \times \vec{E}_T$

$$\omega = v_1 k_i = v_1 k_R = v_2 k_T$$

$$\Rightarrow k_i = k_R = \frac{v_2}{v_1} k_T = \frac{n_1}{n_2} k_T$$

$$v = \frac{c}{n}$$

Now (BC) on $\vec{E}_i + \vec{E}_R$ vs $\vec{E}_T$

$$\Rightarrow \quad e^{i\vec{k}_i\vec{r}} = e^{i\vec{k}_R\vec{r}} = e^{i\vec{k}_T\vec{r}} \quad \text{at the interface } z=0$$

$$x k_{ix} + y k_{iy} = x k_{Rx} + y k_{Ry} = x k_{Tx} + y k_{Ty}$$

must hold for any x and y independently.

$$\Rightarrow \begin{cases} \cancel{x} k_{ix} = \cancel{x} k_{Rx} \\ k_{iy} = k_{Ry} \\ k_{ix} = k_{Tx} \\ k_{iy} = k_{Ty} \end{cases}$$

Let $\vec{k}_i$ be in XZ plane so that $k_{iy} = 0 \Rightarrow k_{Ry} = k_{Ty} = 0$

① $\Rightarrow$ wave vectors of all waves are in XZ plane

$\Rightarrow$ <u>plane of incidence</u>.

$$k_{ix} = k_i \sin\theta_i = k_{Rx} = k_R \sin\theta_R$$

② $\Rightarrow \sin\theta_i = \sin\theta_R \Rightarrow \theta_i = \theta_R$ (law of reflection)

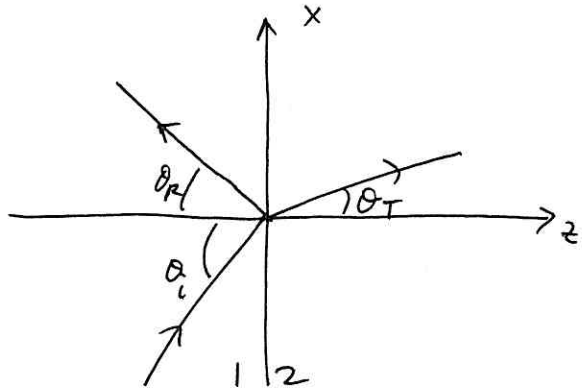
③ $k_i \sin\theta_i = k_T \sin\theta_T \qquad k_i = \frac{\omega}{v_1}$

$$k_T = \frac{\omega}{v_2}$$

$$\frac{\sin\theta_T}{\sin\theta_i} = \frac{k_i}{k_T} = \frac{v_2}{v_1} = \frac{n_1}{n_2} \quad \text{(Snell's law of refraction)}$$

These are three laws of <u>geometric optics</u>!

Consider $\underline{\hat{n} = \hat{y}}$: polarization $\perp$ XZ plane.



$$\vec{k}_i = k_i (\sin\theta_i \hat{x} + \cos\theta_i \hat{z})$$

$$\vec{k}_R = k_R (\sin\theta_R \hat{x} - \cos\theta_R \hat{z})$$

$$\vec{k}_T = k_T (\sin\theta_T \hat{x} + \cos\theta_T \hat{z})$$

$$\vec{B} = \frac{1}{v} (\hat{k} \times \hat{n}) \vec{E}$$

$$\hat{x} \times \hat{y} = \hat{z}$$

$$\hat{z} \times \hat{y} = -\hat{x}$$

$$\vec{E}_i = E_{0i} e^{i(kr-\omega t)} \hat{y}$$

$$\vec{B}_i = \frac{1}{v_1} |\vec{E}_i| (\sin\theta_i \hat{z} - \cos\theta_i \hat{x})$$

$$\vec{E}_R = E_{0R} \hat{y}$$

$$\vec{B}_R = \frac{1}{v_1} |\vec{E}_R| (\sin\theta_R \hat{z} + \cos\theta_R \hat{x})$$

$$\vec{E}_T = E_{0T} \hat{y}$$

$$\vec{B}_T = \frac{1}{v_2} |\vec{E}_T| (\sin\theta_T \hat{z} - \cos\theta_T \hat{x})$$

1) $\varepsilon_1 \underset{\overset{\parallel}{0}}{E_{left,z}} = \varepsilon_2 E_{right,z} = 0$

3) $E_{0i} + E_{0R} = E_{0T}$

2) $\frac{1}{v_1} (E_{0i} + E_{0R}) \sin\theta_i = \frac{1}{v_2} E_{0T} \sin\theta_T$

but $\frac{\sin\theta_T}{\sin\theta_i} = \frac{v_2}{v_1}$

$\Rightarrow E_{0i} + E_{0R} = E_{0T}$ which is same as (3).

4) $\frac{1}{\mu_1} \frac{1}{v_1} (-E_{0i} + E_{0R}) \cos\theta_i = \frac{1}{\mu_2} \frac{1}{v_2} E_{0T} (-\cos\theta_T)$

$$E_{0i} - E_{0R} = \frac{\mu_1 v_1}{\mu_2 v_2} \frac{\cos\theta_T}{\cos\theta_i} E_{0T}$$

let $\alpha = \frac{\cos\theta_T}{\cos\theta_i} > 0$, $\beta = \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1} > 0$

$$\Rightarrow E_{0i} - E_{0R} = \alpha\beta E_{0T}$$

$$2E_{0i} = (1+\alpha\beta) E_{0T} \quad \Rightarrow E_{0T} = \frac{2}{1+\alpha\beta} E_{0i}$$

$$2E_{0R} = (1-\alpha\beta) E_{0T}$$

$$E_{0R} = \cancel{\frac{1-\alpha\beta}{1+\alpha\beta}} \quad E_{0T} - E_{0i} = \left(\frac{2}{1+\alpha\beta} - 1\right) E_{0i} =$$

$$= \frac{1-\alpha\beta}{1+\alpha\beta} E_{0i}$$

Hence E_{0T} is in phase with E_{0i} : $E_{0T} = \frac{2}{1+\alpha\beta} E_{0i}$

while reflected $E_{0R} = \frac{1-\alpha\beta}{1+\alpha\beta} E_{0i}$: $\begin{cases} \text{in-phase for } \alpha\beta < 1 \\ \text{out of phase for } \alpha\beta > 1 \end{cases}$

Assume $\mu_1 \approx \mu_2 \approx \mu_0 \Rightarrow \beta = \frac{n_2}{n_1} = \frac{v_1}{v_2}$

$$\alpha = \frac{\cos\theta_T}{\cos\theta_i}$$

$$\frac{E_{0R}}{E_{0i}} = \frac{1 - \frac{n_2}{n_1}\frac{\cos\theta_T}{\cos\theta_i}}{1 + \frac{n_2}{n_1}\frac{\cos\theta_T}{\cos\theta_i}} = \frac{n_1\cos\theta_i - n_2\cos\theta_T}{n_1\cos\theta_i + n_2\cos\theta_T}$$

<u>Sign?</u> $\frac{n_2}{n_1} = \frac{\sin\theta_i}{\sin\theta_T} \Rightarrow \frac{1 - \frac{\tan\theta_i}{\tan\theta_T}}{1 + \frac{\tan\theta_i}{\tan\theta_T}} = \frac{\tan\theta_T - \tan\theta_i}{\tan\theta_T + \tan\theta_i}$

For $\alpha\beta > 1 \Rightarrow \dfrac{\cos\theta_T}{\cos\theta_i}\dfrac{v_1}{v_2} > 1$

$$\frac{1-\sin^2\theta_T}{1-\sin^2\theta_i} \gtrless \frac{v_2^2}{v_1^2}$$

But $\sin\theta_T = \sin\theta_i \dfrac{v_2}{v_1}$

$$\Rightarrow \quad 1-\sin^2\theta_i\left(\frac{v_2}{v_1}\right)^2 \gtrless \left(1-\sin^2\theta_i\right)\left(\frac{v_2}{v_1}\right)^2$$

$$\Rightarrow \quad 1 \geq \left(\frac{v_2}{v_1}\right)^2 \Rightarrow \frac{v_2}{v_1} < 1 \Rightarrow \frac{n_1}{n_2} < 1 \Rightarrow \boxed{n_2 > n_1}$$

$\Rightarrow \dfrac{E_{0R}}{E_{0i}} < 0$ for $n_2 > n_1$ $\left\{\begin{array}{l}\text{reflected wave is shifted} \\ \text{by } \pi \text{ from } \underline{\text{denser}} \text{ medium}\end{array}\right.$

$\dfrac{E_{0R}}{E_{0i}} > 0$ for $n_2 < n_1$ $\left\{\begin{array}{l}\text{no shift ~~from~~ on going} \\ \text{into optically lighter} \\ \text{media.}\end{array}\right.$

(from averaging over period)

Energy transmission: $I_i = \frac{1}{2}\varepsilon_1 v_1 E_{0i}^2 \cos\theta_i$

$$I_R = \frac{1}{2}\varepsilon_1 v_1 E_{0R}^2 \cos\theta_R$$

$$I_T = \frac{1}{2}\varepsilon_2 v_2 E_{0T}^2 \cos\theta_T$$

$$R = \frac{I_R}{I_i} = \left(\frac{E_{0R}}{E_{0i}}\right)^2 = \left(\frac{1-\alpha\beta}{1+\alpha\beta}\right)^2;$$

$$T = \frac{I_T}{I_i} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1}\left(\frac{E_{0T}}{E_{0i}}\right)^2 \frac{\cos\theta_T}{\cos\theta_i}$$

But $\dfrac{\varepsilon_2 v_2}{\varepsilon_1 v_1} = \dfrac{\varepsilon_2}{n_2}\dfrac{n_1}{\varepsilon_1} = \dfrac{\varepsilon_2}{\sqrt{\varepsilon_2\mu_2}}\dfrac{\sqrt{\varepsilon_1\mu_1}}{\varepsilon_1} = \sqrt{\dfrac{\varepsilon_2}{\mu_2}}\sqrt{\dfrac{\mu_1}{\varepsilon_1}}$

$$v = \frac{c}{n}, \quad n_1 = \sqrt{\frac{\varepsilon_1\mu_1}{\varepsilon_0\mu_0}}$$

while $\beta = \dfrac{\mu_1 v_1}{\mu_2 v_2} = \dfrac{\mu_1}{n_1}\dfrac{n_2}{\mu_2} = \dfrac{\mu_1}{\sqrt{\varepsilon_1\mu_1}}\dfrac{\sqrt{\varepsilon_2\mu_2}}{\mu_2} = \sqrt{\dfrac{\mu_1}{\varepsilon_1}}\sqrt{\dfrac{\varepsilon_2}{\mu_2}}$

$$\Rightarrow \beta = \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1}$$

$$\Rightarrow T = \alpha\beta\left(\frac{2}{1+\alpha\beta}\right)^2 = \frac{4\alpha\beta}{(1+\alpha\beta)^2}$$

$$R = \frac{(1-\alpha\beta)^2}{(1+\alpha\beta)^2} \Rightarrow R + T = \frac{1-2\alpha\beta+\alpha^2\beta^2+4\alpha\beta}{(1+\alpha\beta)^2} = \underline{1}.$$